Chapter 6 • Trigonometry

Solved Exercise 6.4

1. Find each exact value without a calculator.

(i) $\sin 30° = \frac{1}{2}$

(ii) $\cos 30° = \frac{\sqrt{3}}{2}$

(iii) $\tan(\pi/6) = \tan 30°$
$= \frac{1}{\sqrt{3}}$

(iv) $\tan 60° = \sqrt{3}$

(v) $\sec 60° = 2$

(vi) $\cos(\pi/3) = \cos 60°$
$= \frac{1}{2}$

(vii) $\cot 60° = \frac{1}{\sqrt{3}}$

(viii) $\sin 60° = \frac{\sqrt{3}}{2}$

(ix) $\sec 30° = \frac{2}{\sqrt{3}}$

(x) $\operatorname{cosec} 30° = 2$

(xi) $\sin 45° = \frac{1}{\sqrt{2}}$

(xii) $\cos(\pi/4) = \cos 45°$
$= \frac{1}{\sqrt{2}}$

2; (xi) $\frac{1}{\sqrt{2}}$; (xii) $\frac{1}{\sqrt{2}}$

2. Evaluate the eight expressions.

(i) $2(\sqrt{3}/2)(1/2) = \frac{\sqrt{3}}{2}$

(ii) $2(\sqrt{3}/2)(1/2) = \frac{\sqrt{3}}{2}$

(iii) $\frac{2}{\sqrt{2}} \frac{+2}{\sqrt{2}} = 2\sqrt{2}.$

(iv) $\sin(60°+30°) = \sin 90°$
$= 1$

(v) $\cos(60°+30°) = \cos 90°$
$= 0.$

(vi) $\sin(60°-30°) = \sin 30°$
$= \frac{1}{2}$

(vii) $\cos(60°-30°) = \cos 30°$
$= \frac{\sqrt{3}}{2}$

(viii) $\tan(\pi/6)\cot(\pi/6)+1 = 1+1$
$= 2.$

3. Given $\sin(\frac{\pi}{4}) = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}}$, evaluate the three expressions.

(i) $2\sin 45° - 2\cos 45° = \frac{2}{\sqrt{2}} \frac{-2}{\sqrt{2}}$
$= 0.$

(ii) $3\cos 45° + 4\sin 45° = \frac{3}{\sqrt{2}} \frac{+4}{\sqrt{2}}$
$= \frac{7}{\sqrt{2}}$
$= \frac{7\sqrt{2}}{2}$

(iii) $5\cos 45° - 3\sin 45° = \frac{5}{\sqrt{2}} \frac{-3}{\sqrt{2}}$
$= \frac{2}{\sqrt{2}}$
$= \sqrt{2}.$

studysapiens.com